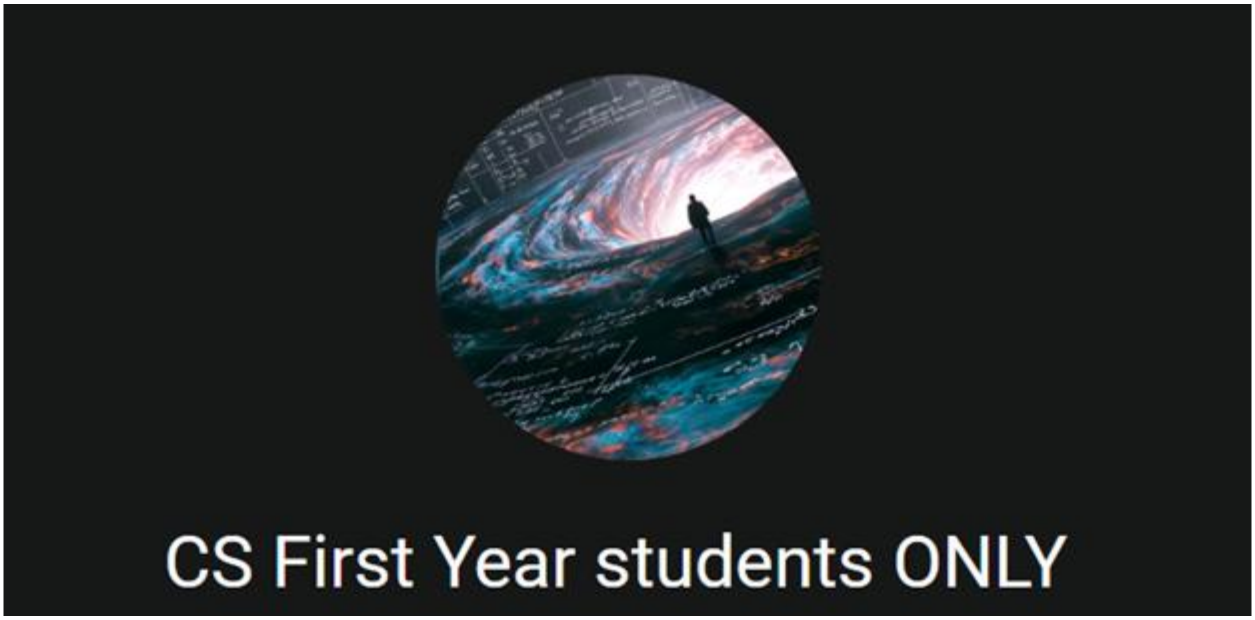
Final Exam For Second Semester (2024/2025)




Leader's :

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Ministry of High Education
Obour High Institute for Management & Informatics

# Final Exam for second Semester (2024/2025)
## Model (A)

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| Course Name: Linear Algebra | Course Code: BS 102 Bylaw: 2024 |
| Year: First | Department: Computer Science |
| Date of Exam: Saturday 31/5/2025 | Time Allowed: 2 hour Total Points: 60 |
| Course Instructor: Dr. Dina Ezzat Mahmoud | No. of questions: (2) No. of pages: (2) |

Exam instructions

- **Make sure that the question form is identical to the answer sheet form**
- **The answer sheet must be in good condition from the moment received until it is delivered after taking the exam**
- **Do not bend the answer sheet or lean on it while answering**

Answer the following questions:

**Question One: [18 points]** (ILOs: a1, b1, c2)

a. Choose the correct answer from (A), (B), (C) or (D), to complete the sentence:

1- Which one of the following is elementary matrix

a- $\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & a \end{bmatrix}$ b- $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ c- $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ d- $\begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$ ✓

2- When multiplying 3 by the matrix $\begin{bmatrix} 1 & -2 & 2 \\ 3 & 4 & 1 \\ 2 & -1 & 1 \end{bmatrix}$

a- $\begin{bmatrix} 3 & -6 & 6 \\ 9 & 12 & 3 \\ 6 & -3 & 3 \end{bmatrix}$ b- $\begin{bmatrix} 3 & -6 & 6 \\ 3 & 4 & 1 \\ 2 & -1 & 1 \end{bmatrix}$ c- $\begin{bmatrix} 1 & -2 & 2 \\ 9 & 12 & 3 \\ 2 & -1 & 1 \end{bmatrix}$ d- $\begin{bmatrix} 1 & -2 & 2 \\ 3 & 4 & 1 \\ 6 & -3 & 3 \end{bmatrix}$

3- From the following the linear equation

a- $3x + y = 5$ b- $3x + e^3y + z = 0$ c- $\sqrt{2x} + y = 0$ d- $x^2 - y = 6$ ✓

4- The transpose of the matrix $A = \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 3 & 0 \end{bmatrix}$

a- $\begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ b- $\begin{bmatrix} 1 & 3 \\ -1 & 2 \\ 0 & 3 \end{bmatrix}$ c- $\begin{bmatrix} 1 & -1 & 0 \\ 3 & 2 & 3 \end{bmatrix}$ d- $\begin{bmatrix} -3 & -1 \\ -2 & -1 \\ -3 & 0 \end{bmatrix}$

5- $A^3$ for $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} = \cdots\ldots$

a- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ b- $\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$ c- $\begin{bmatrix} 1 & 8 \\ 0 & -1 \end{bmatrix}$ d- $\begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$

6- The vectors in $R^2$ that are orthogonal to $u = (3,1)$ has the form

a- $(1,-3)$ b- $(-3t,t)$ c- $(-3,1)$ d- $(t,-3t)$

b. Put true or false

1. the subset $w = (x_1, x_2, x_3)$ is not a subspace of $R^3$, Where $x_1, x_2, x_3$ are real numbers. ✗
2. $(9,10,13)$ can be written as a linear combination of $(1,2,1),(2,1,3),(3,4,5)$ ✓

- If A and B are matrices then

3. $(A^T)^T = A^{-1}$
4. $(A+B)^T = A^T + B^T$

Question Two: [42 points] (ILOs: a1, b1, c2)

a. Show that the set A={(1,2,3), (0,1,2),(-2,0,1)} span $R^3$

b. Determine whether set a linearly dependent or linearly independent. (use row echeln form)

c. Use Gauss-Jordan elimination to find the inverse of the matrix $B = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 7 & 9 \\ -1 & -4 & -7 \end{bmatrix}$

d. Find an equation of the line passing through the points (1,2) and (-2,3)

e. Use Cramer's rule to solve the system of linear equation:

$$2x + y - z = 3$$
$$x - 2y + 3z = -4$$
$$3x + y + 2z = 5$$

f. Given the matrix A=$\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, find the eigenvalues and the corresponding eigenvectors.

End of Questions
Good Luck,,,

Examiners:

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